Table of Contents

Software Development Plan

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Conforms to	ISO/IEC/IEEE 16326:2019

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1. Overview

1.1 Project summary

Purpose and objectives. Deliver NOVA, a locally hosted agent converting software requirements into structured test plans and retaining user corrections as durable memory. Objectives are stated in NOVA-VS-001 section 1.3.

Scope. As defined in NOVA-VS-001 section 3.

Assumptions and constraints.

ID	Constraint	Effect
CON-1	Approximately 130 to 180 engineering hours	Scope reduction order pre-committed
CON-2	Single engineer, no external review	Automated gates substitute for peer review where possible
CON-3	Local inference, 6 GB VRAM	Model class fixed at 4B parameters. Confirmed by measurement
CON-4	Local deployment only	Container composition is the deployment artifact
CON-5	Synthetic and open-source data only	Evaluation data is committable
CON-6	No model training	Adaptation through retrieval and configuration only

Deliverables.

Deliverable	Form
NOVA application	Source, containerized, startable from a clean checkout
Document set	NOVA-VS-001, SRS-001, SAD-001, STP-001, TM-001, SDP-001, RR-001, DL-001, ADRs
Evaluation dataset	Content-hash-versioned, committed
Automated test suites	Source, gated in continuous integration
Spike reports	Committed with reproducible harnesses and raw output
Demonstration recording	Video

Schedule and effort summary. Approximately 165 hours across seven milestones at 10 to 15 hours per week.

Availability	Elapsed duration	Assessment
15 h/week	~11 weeks	Within budget
12.5 h/week	~13 weeks	Target
10 h/week	~16.5 weeks	Over budget. The first two scope reductions become mandatory

1.2 Evolution of this plan

Effort figures are estimates, not measurements, and will prove inaccurate. **Milestone exit criteria govern, not hours consumed.** A milestone completes when its criteria are met.

This plan is revised at each milestone boundary. Revisions are recorded in the revision history rather than applied silently.

2. References

Reference	Title
NOVA-VS-001	Vision and Scope
NOVA-SRS-001	Software Requirements Specification
NOVA-SAD-001	Software Architecture Document
NOVA-STP-001	Software Test Plan
NOVA-TM-001	Threat Model
NOVA-RR-001	Risk Register
NOVA-DL-001	Decision Log

3. Definitions

See NOVA-SRS-001 section 1.4.

4. Project organization

4.1 External interfaces

No external organization participates. The project has no sponsor, customer, or supplier.

4.2 Internal structure and responsibilities

Role	Held by	Responsibility
Project lead	Laxmi Poudel	Scope, schedule, acceptance, all decisions
Architect	Laxmi Poudel	Architecture, decision records
Developer	Laxmi Poudel	Implementation
Test manager	Laxmi Poudel	Test plan, gate definitions, thresholds
Operator	Laxmi Poudel	Deployment, model provisioning, backup

Single-engineer project. The consequent absence of independent review is recorded as risk R-17 and constraint CON-2, and is stated rather than concealed.

5. Managerial process plans

5.1 Start-up plan

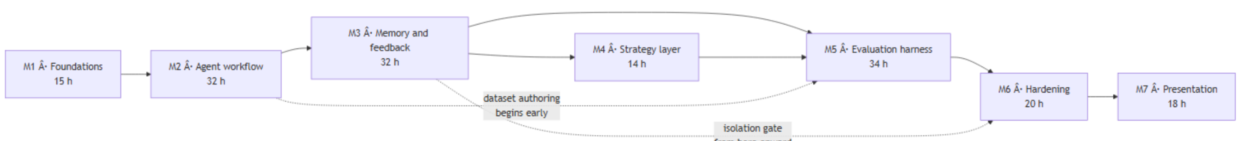
Effort estimation. Bottom-up by milestone, derived from the vertical slices in section 6.2. Estimates carry no historical basis and are expected to be inaccurate in both directions.

Resource acquisition. No procurement. Software dependencies are open-source. Hardware is existing.

Entry criteria for implementation. Implementation begins when the items in section 5.6 are resolved.

5.2 Work plan

Milestones are sequenced by technical prerequisite and by risk reduction, not by feature appeal. The highest-uncertainty assumptions are deliberately addressed first, because discovering a failure late would be unrecoverable.



M1. Foundations, 15 h

Partially complete.

Objective	Eliminate the assumptions capable of invalidating the plan. Fix the decisions that are expensive to reverse
Scope	Model selection spike; embedding spike; repository skeleton; datastore with vector extension and migrations; inference gateway with a fake implementation; deployment composition; continuous integration skeleton
Exit criteria	A selected model produces schema-conformant plans at measured latency; embedding model and vector dimension fixed; container-to-host inference reachability verified
Complete	Model selection spike, partial. Three of five candidates measured, harness and raw output committed
Outstanding	Embedding spike (blocking); remaining model measurements; deployment composition; migrations; gateway; continuous integration

M2. Agent workflow, 32 h

Objective	End-to-end task execution without memory or adaptation. The substrate on which everything else depends
Scope	Task entity and state machine; job table and worker; orchestrator; generator; deterministic checks; trace persistence; minimal interface; four playbooks as versioned data
Exit criteria	A requirement submitted through the interface returns a schema-conformant plan; the trace exposes every step; a terminated worker's job is reclaimed with its partial trace intact
Concurrent	Evaluation dataset authoring begins, extending the twenty cases produced during the model spike
Resolve before M3	Test plan schema frozen at version 1; playbook definitions stable enough to scope memory against

M3. Memory and feedback, 32 h

The milestone the project exists to deliver.

Objective	Close the feedback loop
Scope	Memory entity and vector index; embedding on confirmation; scored retrieval; feedback events and normalization; extraction with the validation screen; confirmation flow; conflict detection; memory management interface; applied-lessons presentation
Exit criteria	A correction recorded against task N demonstrably alters task N+1; deleting the memory reverts the behaviour; the isolation suite is green and gating
Definition of	Extraction precision measured against approximately 15 hand-labelled authentic

done	corrections; deletion completeness proven by test
Principal risk	R-01. Extraction precision is the highest uncertainty in the project

Mid-milestone checkpoint at approximately 15 hours. Extraction precision is measured before the surrounding interface is built. Should a local model prove unable to derive usable rules, the fallback is user-authored rules with model assistance: a smaller capability, still honest, and far cheaper to adopt at that point than after dependent interface work.

M4. Strategy layer, 14 h

Objective	Make playbook selection responsive to recorded outcomes, within guardrails
Scope	Outcome score entity; bounded exploration; cold-start defaults; exploration labelling in the interface; score recomputation from the event log
Exit criteria	Selection demonstrably shifts following recorded outcomes; exploration remains within bound; scores rebuild exactly from the event log

The smallest milestone, off the critical path, and consequently the first candidate for reduction.

M5. Evaluation harness, 34 h

Objective	Render quality measurable and regression unmergeable
Scope	Dataset extension to approximately 50 stratified cases (15 h in itself); metric computation; adversarial suite; configuration versioning with the activation gate; continuous integration evaluation job; results presentation
Exit criteria	A deliberately degraded prompt version is blocked by the gate
Definition of done	Run-over-run variance measured **before** thresholds are set, with thresholds justified in writing by that variance; adversarial suite green; Correction Recurrence Rate measured across at least two runs
Principal risks	R-03 nondeterminism destabilizing the gate; R-02 dataset effort underestimated

The largest milestone and the most likely to overrun, the dataset being the cause. It also produces the artifact of greatest evidential value, so reduction here costs more per hour saved than elsewhere.

M6. Hardening, 20 h

Objective	Render the system operable and demonstrate its security properties
Scope	Distributed tracing; cost and duration presentation; structured logging with content-

	absence assertions; degraded-mode behaviours; health and readiness including provider reachability; secret scanning and dependency audit; backup and restore drill; clean-checkout deployment verification
Exit criteria	Every degraded mode demonstrable; no content present in log output, asserted by test; deployment succeeds from a clean checkout on a separate host

M7. Presentation, 18 h

Objective	Render the work legible to a reader with limited time
Scope	Repository README; document set finalization; case study; demonstration recording; screenshots; resolution of every provisional value
Exit criteria	Every provisional value is either replaced with a measurement or the associated claim is removed
Principal risk	R-07. The pressure to substitute optimistic figures peaks here. The claim discipline in section 7.6 is reviewed at the start of this milestone, not at its end

5.3 Control plan

Requirements control. The SRS is the baseline. Changes are made to the SRS first, then implemented. Scope additions require a corresponding reduction elsewhere or an explicit budget revision.

Schedule control. Progress is measured by milestone exit criteria, not by hours consumed or tasks closed.

Metrics collection. Product and process metrics are defined in NOVA-STP-001 section 5.5.

Reporting. Self-directed. Milestone completion is recorded in this document's revision history.

5.4 Risk management plan

Risks are recorded in NOVA-RR-001 and reviewed at each milestone boundary rather than continuously.

A risk closes only when evidence closes it. A planned mitigation does not close a risk, and neither does elapsed time.

Two entries are prerequisites rather than ordinary risks. Dependent work does not begin until they are resolved, because the cost of proceeding while wrong is rework rather than delay:

ID	Prerequisite
R-04	Embedding model and vector dimension selected. Blocks schema definition
R-01	Extraction precision measured. Blocks dependent interface work in M3

Pre-committed scope reduction order. Decided in advance so that reduction proceeds from a plan rather than under pressure. The decision point is week 6.

1. Evaluation results interface reduced to a command-line report and a committed summary
2. Playbooks reduced from four to two, preserving the strategy layer while halving the work
3. Hosted inference provider removed, retaining local only. Costs the demonstration fallback
4. Memory edit and pin reduced to view and delete
5. Evaluation dataset reduced from approximately 50 to approximately 25 cases
6. Cost and duration presentation moved from the interface to logs

Never reduced: the feedback loop, the isolation suite, and the evaluation regression gate. These three constitute the project. Removing any one leaves an unremarkable generation wrapper.

5.5 Closeout plan

The project closes when every requirement of priority Must has passed its verification method, the acceptance criteria in section 6.4 are met, and the document set is finalized with no unresolved provisional value.

5.6 Entry criteria for implementation

Assessment against the current state of the document set.

#	Criterion	State	Evidence or gap
1	Problem defined	Complete	NOVA-VS-001 Â§1
2	Users and operational scenarios documented	Complete	NOVA-SRS-001 Â§1.3.3, Â§3.5
3	Objectives and success measures defined	Complete	NOVA-VS-001 Â§1.3, NOVA-STP-001 Â§5.5
4	Scope, assumptions, constraints, exclusions documented	Complete	NOVA-VS-001 Â§3
5	Functional requirements specified	Complete	NOVA-SRS-001 Â§3.1
6	Non-functional requirements specified	Complete	NOVA-SRS-001 Â§3.2 to Â§3.12
7	User interface design available	**Gap**	Operational scenarios exist. No screen inventory or wireframes. The trace viewer and applied-lessons presentation carry the project's explanatory value and neither is designed
8	Architecture documented and reviewed	Partial	NOVA-SAD-001 complete. Reviewed by the author only, per CON-2

#	Criterion	State	Evidence or gap
9	Alternatives and trade-offs documented	Complete	ADRs, NOVA-DL-001
10	Data model, ownership, retention, migration understood	**Blocked**	Entities and retention defined. The vector dimension is unselected and is fixed at schema creation
11	Interface contracts defined	Complete	NOVA-SRS-001 Â§3.4
12	Security and privacy requirements identified	Complete	NOVA-TM-001
13	Dependencies, risks, open items tracked	Complete	NOVA-RR-001
14	Test strategy documented	Complete	NOVA-STP-001
15	Observability requirements documented	Complete	NOVA-SAD-001 Â§8.7
16	Deployment, rollout, rollback, support documented	Complete	Section 6.3 and NOVA-SAD-001 Â§7
17	Work decomposed into estimable, demonstrable increments	Complete	Section 6.2
18	Acceptance criteria defined	Complete	Section 6.4

Fourteen complete, two partial or blocked, two gaps.

Outstanding items, in order of consequence.

ID	Item	Effort	Rationale
G1	Select the embedding model and fix the vector dimension	~4 h	**Blocking.** Selecting after schema creation requires re-embedding every memory and rebuilding the index
G2	Complete the model selection spike across all cases and candidates	~3 h, largely unattended	Strongly recommended. A provisional model choice is revisable without rework, unlike the vector dimension
G3	Produce a screen inventory and wireframes for the four views	~4 h	Required before interface work. Does not block foundation or backend work
G4	Verify container-to-host inference reachability and datastore provisioning	~1 h	Removes the most probable early impediment

G1 and G4 together are approximately one working day and are the practical entry condition.

6. Technical process plans

6.1 Process model

Incremental delivery through vertical slices, each independently demonstrable. No fixed iteration length. Work is gated by milestone exit criteria rather than by time-boxing, which suits a single-engineer project with variable weekly availability.

6.2 Vertical slices

Decomposition is by demonstrable outcome rather than by architectural layer.

ID	Slice	Milestone	Demonstrable outcome
S1	Requirement to plan	M2	A requirement submitted through the interface returns a schema-conformant plan
S2	Durable asynchronous execution	M2	The worker is terminated mid-task; the job is reclaimed and the partial trace survives
S3	Trace inspection	M2	Every step, duration, and token count of a completed task is readable
S4	Feedback capture	M3	Accept, edit, and reject recorded against individual cases
S5	Correction to memory	M3	A correction produces a candidate; confirmation stores a memory with provenance
S6	Memory application	M3	A later similar requirement retrieves and applies the memory, with its score displayed
S7	Memory reversal	M3	Deleting the memory and re-running removes the resulting behaviour
S8	Conflict surfacing	M3	Two contradictory rules are detected and presented rather than silently resolved
S9	Adaptive selection	M4	Selection shifts following recorded outcomes; scores rebuild from the event log
S10	Quality measurement	M5	An evaluation run against the versioned dataset reports every metric
S11	Regression prevention	M5	A degraded prompt version fails the gate
S12	Adversarial resilience	M5	Injection, poisoning, and isolation scenarios all pass
S13	Operability	M6	Degraded modes behave as specified; clean-checkout deployment succeeds; backup restores
S14	Legibility	M7	Demonstration, screenshots, finalized documents, measured figures

S6 and S7 together constitute the primary demonstration. S7 establishes causation rather than asserting it. S11 is the evidence that quality control is enforced rather than described.

Entry criteria for a slice.

- Acceptance criteria written and verifiable
- Data read and written is defined in the data model
- Failure modes named with intended behaviour for each
- Every dependent decision either recorded or holding an explicit default
- Demonstrable without waiting on a later slice

Exit criteria for a slice.

- Acceptance criteria pass
- Unit tests cover the non-trivial logic introduced
- Any new data access method is tenant-scoped and carries an isolation test
- Failure paths exercised, not only the nominal path
- Trace or logging sufficient for diagnosis at a remove of months
- Documentation affected by the change updated in the same commit
- No claim introduced without supporting evidence

6.3 Infrastructure plan

Concern	Approach
Development environment	Local workstation. Containerized datastore, host-resident inference runtime
Continuous integration	Hosted runners. Ephemeral datastore container. Fake inference provider for integration; pinned model for evaluation
Deployment	Container composition, startable from a clean checkout by one command. Inference runtime provisioned on the host, documented separately
Rollback	Version control revert plus a documented dump and restore. The restore procedure is drilled at least once, because an untested backup is not a backup
Support	Self-supported. Defects recorded as repository issues

6.4 Product acceptance plan

Acceptance requires every criterion below to be demonstrable end to end.

ID	Criterion	Traces to
AC-1	A requirement submitted through the interface returns a schema-conformant plan	FR-1, FR-2
AC-2	The trace exposes retrieved memories with scores, model calls, tokens, and	FR-8

ID	Criterion	Traces to
	durations	
AC-3	A rejection with a freeform correction produces a candidate memory for confirmation	FR-5, FR-6
AC-4	A later structurally similar requirement retrieves and applies that memory without restatement	FR-4
AC-5	Memory browse, edit, pin, and delete all function. Deletion prevents subsequent retrieval	FR-7
AC-6	Two conflicting memories are detected and presented rather than silently resolved	FR-6.5
AC-7	The evaluation suite executes against the committed dataset and reports every metric	FR-9
AC-8	**A deliberately degraded prompt version is blocked by the regression gate**	FR-10.1
AC-9	Selection demonstrably shifts following recorded outcomes, with exploration remaining bounded	FR-3
AC-10	The adversarial suite passes: injection, poisoning, isolation	NFR-15 to NFR-22
AC-11	Deployment from a clean checkout produces a working system by a single command	NFR-32
AC-12	The inference provider can be changed without code modification	FR-11
AC-13	The Correction Recurrence Rate is measured and reported across at least two runs, including any deterioration	FR-9.2

AC-8 carries the greatest evidential weight. A gate that demonstrably blocks a degrading change is the distinction between an enforced quality system and a described one.

7. Supporting process plans

7.1 Configuration management

Item	Control
Source and documents	Version control. Documents authored in Markdown alongside the code, reviewed through the same mechanism
Distributable documents	Generated from Markdown into DOCX and PDF. Generated artifacts are never edited directly
Database schema	Forward-only versioned migrations
Prompts and configuration	Versioned records in the datastore, activation gated on evaluation

Item	Control
Playbooks	Versioned repository data, changed through review and the evaluation gate
Evaluation dataset	Content-hash versioned. Any edit produces a new version
Dependencies	Pinned with a lockfile. Base images pinned by digest

Document status values, in lifecycle order: `Draft`, `Review`, `Approved`, `Active`, `Superseded`. Versions are `MAJOR.MINOR`; minor for edits within a status, major on approval.

7.2 Verification and validation

Defined in NOVA-STP-001. Requirement-to-verification traceability is in NOVA-SRS-001 section 4.

7.3 Documentation plan

Document	Author	Reviewed at
NOVA-VS-001 Vision and Scope	Laxmi Poudel	Prior to M1 completion
NOVA-SRS-001 SRS	Laxmi Poudel	Prior to M2
NOVA-SAD-001 SAD	Laxmi Poudel	Updated at every milestone boundary
NOVA-STP-001 Test Plan	Laxmi Poudel	Prior to M2, updated at M5
NOVA-TM-001 Threat Model	Laxmi Poudel	Prior to M3, updated at M5
NOVA-SDP-001 SDP	Laxmi Poudel	Every milestone boundary
ADRs	Laxmi Poudel	At the point of decision
Spike reports	Laxmi Poudel	On completion of each spike

Architecture decision records are written when the decision is made, not retrospectively. Retrospective records read as reconstruction and an experienced reviewer will recognize them.

Work suited to low-availability sessions, none of which blocks other work: evaluation dataset authoring from M2 onward; decision records as decisions occur; documentation drafting; screenshot capture as capability lands, so that M7 is not compressed.

7.4 Quality assurance

Gate	Enforcement
Static analysis and type checking	Blocks merge
Unit tests with coverage floor on core logic	Blocks merge
Integration tests	Blocks merge
Interface description drift	Blocks merge
Adversarial suite	Blocks merge unconditionally
Cross-tenant isolation	Blocks merge unconditionally

Gate	Enforcement
Evaluation regression gate	Blocks merge on prompt, playbook, or configuration change

Thresholds derive from measured variance rather than from intent. A gate that flaps is disabled in practice, and a disabled gate is worse than none because it presents as protection.

7.5 Reviews and audits

No independent reviewer is available. Substitutes:

- Automated gates as the primary review mechanism
- A documentation accuracy pass at M7, tracing every stated claim to a test, a metric, or a decision record
- Charter-based exploratory sessions, with findings recorded as issues rather than corrected silently
- One external reader for document comprehension at M7

7.6 Claim discipline

No performance, quality, or improvement figure appears in any document, repository README, or external material until it has been measured and can be reproduced. Unmeasured values appear as bracketed placeholders.

A placeholder that cannot be filled becomes a removed claim, not a softened one.

Measured figures are stated with their method, sample size, and reference configuration. Figures without that qualification are not reproducible and are therefore not claims.

Until a trained model exists and has been measured, all material states that the system does not fine-tune, in the present tense. The roadmap may record fine-tuning as planned; a capability statement may not.

7.7 Problem resolution

Defect severity classification and response are defined in NOVA-STP-001 section 9. Every defect of severity S1 or S2 receives a failing test before the corrective change.

7.8 Process improvement

Effort estimates are compared against actuals at each milestone boundary and the variance recorded, so that later estimates carry some historical basis rather than none.

8. Post-release roadmap

Ordered by anticipated value relative to effort. Not committed.

#	Capability	Note
1	Executable test code generation	Highest perceived value, largest scope increase
2	Multi-user authentication	Schema is prepared. Adds substantive backend and security depth
3	Defect report structuring as a second task type	Approximately 80% infrastructure reuse. Demonstrates architectural generality
4	Issue tracker ingestion	Removes manual transfer
5	Team-shared memory with review	Requires item 2
6	Hosted deployment	Only if a publicly reachable instance becomes worthwhile
7	Model fine-tuning	Prerequisites below

On item 7. The initial release produces every input a fine-tune would require, which is why it is sequenced last rather than excluded.

Prerequisite	Source
A labelled dataset of sufficient volume	Accumulated feedback events and confirmed corrections. The initial release is the collection phase
A measured baseline	Evaluation metrics across at least two runs
A held-out set never used in training	The dataset split, fixed and documented before any training
A reproducible training configuration	Pinned base model, seed, hyperparameters, dataset version hash
Sufficient hardware headroom	Requires its own spike. 6 GB is constraining, and sequence length and batch size will bind

Should the trained model fail to exceed the retrieval-based baseline, that is a legitimate finding and is published as such. Small-model fine-tuning frequently loses to effective retrieval. A negative result measured correctly is a stronger artifact than a positive result that cannot be defended.

None of this permits describing the system as self-training before a trained model exists.

Revision history

Version	Date	Author	Change
0.1	2026-09-08	Laxmi Poudel	Initial draft